

VAIBHAV SHARMA

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github.com/sharmavaibhav31 🌐 [Portfolio - vaiibhavsharmaa.me](https://Portfolio-vaiibhavsharmaa.me)

SUMMARY

Backend developer who builds the behind-the-scenes systems that apps depend on, and tests them until they break. At a startup, built an AI feature that turns plain-English requests into automated tasks, fixed a bug that was silently stalling customer workflows, and found three security flaws during testing. Also lead an open-source project with 75+ outside contributions reviewed, and run my college coders club, including a hackathon with 880+ participants.

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, TypeScript/JavaScript
Backend & APIs: Spring Boot, Spring Security, FastAPI, Flask, Node.js, Express, REST API Design, JWT, RBAC
Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis Streams, Flyway, Prisma
Security & Systems: Linux, Seccomp, Namespaces, Capability Dropping, rlimits, strace, Security Testing (IDOR, Timing Attacks)
DevOps & Testing: Docker, Nginx, AWS EC2, GitHub Actions, Git, Testcontainers, pytest, Contract/E2E Testing
AI & Frontend: LLM Integration (Gemini, llama.cpp, Ollama), React, Next.js, Flutter

EXPERIENCE

Software Developer Intern

Jun 2026 – Sep 2026

LazyStudents.in · Next.js, TypeScript, PostgreSQL, Redis, Docker, Gemini API

Remote

- Built **Lazy Command**, an AI feature that turns plain-English requests into automated multi-step workflows across **100+ platform tools**.
- Found why customer workflows were silently stalling mid-run (four independent causes) and shipped a recovery job that safely restarts stuck runs, even with multiple workers active.
- Rebuilt a broken webhook inspection tool end-to-end. In a separate security audit of the platform, identified an access-control flaw (IDOR), a login timing leak, and a build-time authorization bypass.
- Led a 5-member intern team, tested **65+ production tools**, and reviewed **30+ pull requests**; set up the team's Git workflow (named branches, Conventional Commits, rebase policy).

AI/ML Backend Intern

Sep 2025 – Nov 2025

Infosys Springboard · Python, Flask, MongoDB, MusicGen, YAMNet

Virtual

- Built the backend for an AI music generation platform, coordinating three ML models behind one API; cut startup time **45%** (2.2s → 1.2s) over 500+ requests and added fallback routing so the service keeps working if a model fails.
- Stored song metadata and securely hashed user credentials in MongoDB.

Flutter Developer Intern

Jul 2025 – Sep 2025

ClubChat · Flutter, Dart

Remote

- Shipped event-feed and real-time chat features for a mobile app used by **10,000+ students**.
- Fixed **18 production stability issues** across two releases, including 3 critical crash-on-launch bugs.

PROJECTS

Arachnode — Open-Source Job Discovery Platform | Python, FastAPI, Redis Streams, Docker

[!\[\]\(206536f97fdb267876a3a10ea42b0254_img.jpg\) GitHub](#)

- Built a self-hosted platform that finds job postings across multiple sites, locates recruiter contacts, and drafts outreach emails, automating the job-hunt funnel I was running by hand during placement season.
- Designed as **9 independent services** communicating through Redis Streams (chosen over Kafka for lower operational overhead on a single machine), with a **4-tier automated test suite**: unit, integration with throwaway databases, contract, and end-to-end.
- Open-sourced and maintained with outside contributors through GSSoC and ELUSOC; **26 stars, 47 forks**.

Offline AI Assistant Shell | *C, llama.cpp, Seccomp, Linux Namespaces*

 [GitHub](#)

- Built a command-line shell with an AI assistant that runs **entirely offline** on a laptop CPU (TinyLlama via llama.cpp), so no commands or data leave the machine.
- Contained the AI process with four layers of protection (syscall whitelisting, namespace isolation, dropped privileges, resource limits) and verified it with **strace auditing** and model-integrity checks; runs at 10–20 tokens/s in ~1–1.5GB of memory.

VulnGraph — Commit-Level Security Regression Detector | *Python, FastAPI, React, NetworkX*

- As part of a 4-person team, built a tool that pinpoints **which code commit made an application easier to attack**, which scanners that analyse a single snapshot (SonarQube, CodeQL) don't report.
- Maps the routes from user input to sensitive operations (like database queries) before and after each commit, then scores every commit from 0 to 1 based on new attack routes opened and protections weakened; the score is deterministic, so it can gate CI. Supports Java and Python and runs fully locally.
- Delivered as a web dashboard with a 3D attack-path graph and a VS Code extension that shows the risk score before you commit.

Appraisal Management System | *Java, Spring Boot, PostgreSQL, React*

- Replaced the college's paper-based faculty appraisal process with a workflow system: appraisals move through draft, review, finalisation and lock stages, with **4 user roles** and an audit trail on every change.
- Gathered requirements from college management; now porting it to the college ERP's stack (Node.js, Angular, Keycloak) after the administration asked for integration.

Additional builds: HPMS — a QR-code hostel gate-pass system with five user roles, proposed to college administration for deployment • Academic Timetable Scheduler — generates conflict-free timetables in under 2 minutes by enforcing 8 hard constraints **More projects:** github.com/sharmavaibhav31 — [17]+ public repositories

LEADERSHIP & COMMUNITY

Project Admin — Arachnode

Feb 2026 – Present

GSSoC 2026 · ELUSOC 2026 · Aperture 3.0

- Run Arachnode as an open-source project: reviewed and merged **75+ pull requests** from outside contributors, closed **30+ issues**, and mentored first-time contributors through GSSoC and ELUSOC.
- Set the project's standards: API contract rules, GitHub Actions CI pipelines, and a README template that every service follows.
- Contributed 2 merged pull requests to the GitCanvas project under Aperture 3.0.

President, Coders Club

Oct 2025 – Present

MITE Campus

- Organized **Hack n' Seek**, a hybrid hackathon with **700+ online** and **180+ on-site** participants across multiple colleges; when demand from other colleges outgrew the planned college-only event within hours, re-planned it as a hybrid format.
- Delivered **10+ workshops** on Git, DSA, and backend architecture; mentored **60+ students** individually and in project teams.

EDUCATION

Mangalore Institute of Technology and Engineering

Sep 2023 – Jun 2027

B.E., Computer Science and Engineering; CGPA: 9.37

Karnataka, India

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Cloud Computing, Machine Learning

Competitive Programming: 150+ LeetCode problems • 100+ TUF/Striver SDE Sheet problems

CERTIFICATIONS

Google Cloud: Cloud Security Engineer, Security in Google Cloud, Google Cybersecurity Professional Certificate

Oracle OCI: DevOps Professional, Data Science Professional, Generative AI Professional, Generative AI Foundations Associate